AI-Powered NLP Document Intelligence Platform

Project Case Study | Requirements Management · Cross-Functional Coordination · Sprint Tracking · Production Delivery

Confidentiality: Client and organisation names have been omitted. This case study reflects my actual responsibilities and does not disclose confidential project information.

1. Project Overview

The project involved an AI-powered NLP document intelligence platform that enabled users to ask natural-language questions against document collections and receive answers supported by relevant document references.

The engagement required collaboration between business stakeholders and multiple technical teams to translate requirements into actionable deliverables, coordinate execution, monitor progress and support the platform through testing and production delivery.

The Problem and Context

The client's teams spent hours manually searching more than 5,000 policy, contract and report documents (PDF and Word) to find answers. Results were inconsistent and rarely showed where an answer came from. The business needed a platform that returned accurate answers in seconds, each backed by the document it came from.

Project Snapshot

Project Attribute	Details
Project	AI-Powered NLP Document Intelligence Platform
My Role	Business Analyst – Requirements & Delivery Coordination
Engagement Duration	4 months (8 two-week sprints)
Team Size	20+ members across NLP/AI, backend, QA and DevOps

Project Attribute	Details
Progress Tracking	Excel-based project tracker
Delivery Outcome	Delivered to production on the planned date

2. My Role and Responsibilities

My role focused on connecting business requirements with technical execution and maintaining delivery visibility across the engagement.

Requirements Management

- Gathered and clarified business and functional requirements.
- Prepared Business Requirements Documents (BRDs) and Functional Requirements Documents (FRDs).
- Documented use cases, workflows and expected system behaviour.
- Coordinated requirement clarifications between client stakeholders and internal technical teams.

Cross-Functional Coordination

- Coordinated with NLP/AI, backend development, QA and DevOps teams.
- Facilitated communication between client stakeholders and internal delivery teams.
- Followed up on pending activities and cross-team dependencies.
- Communicated requirement clarifications to relevant technical stakeholders.

Sprint Tracking and Progress Monitoring

- Worked within two-week sprint delivery cycles.
- Collected progress updates from relevant team members.
- Personally maintained and updated task statuses in an Excel-based project tracker.
- Monitored pending activities and dependencies.
- Followed up with relevant teams on delivery blockers.
- Consolidated progress information for stakeholder communication.

Testing and Release Coordination

- Supported testing and validation activities.
- Coordinated around UAT requirements and feedback.
- Supported release-related coordination across development, sandbox and production environments.

3. Project Execution Approach

The engagement ran over eight two-week sprints and followed six stages.

1. **Requirement gathering and analysis:** documented the client's needs in 2 BRDs, 3 FRDs and about 60 requirements and user stories.
2. **Technical team coordination:** passed requirements to the NLP/AI, backend, QA and DevOps teams and tracked clarifications and dependencies.
3. **Sprint execution and tracking:** collected updates each sprint and kept task statuses current in the Excel tracker.
4. **Progress monitoring and follow-up:** chased blockers and consolidated progress for stakeholders.
5. **Testing and validation:** coordinated two UAT cycles with the client.
6. **Production delivery:** supported release coordination across development, sandbox and production environments.

4. Project Tracking and Stakeholder Communication

An Excel tracker gave every team one view of completed, pending and blocked work across all eight sprints. I updated it myself after collecting progress from each team and sent the client a weekly status summary. I also bridged client stakeholders and internal teams so clarifications and updates reached the right people.

The tracker's main value was early warning. In sprint 5 it showed a backend task waiting on the NLP team's output, which let me escalate before it could delay UAT.

5. Key Challenges and How I Handled Them

Coordinating 20+ people across four technical functions meant three situations needed active handling.

1. A requirement changed mid-project

- **Situation:** In sprint 4 the client asked for page-level source references in every answer, beyond the original scope.
- **Action:** I assessed the impact with the NLP and backend leads, updated the FRD and reprioritised the backlog with the client's agreement.
- **Result:** The feature shipped in sprint 6 and the release date did not move.

2. A cross-team dependency blocked progress

- **Situation:** In sprint 5 the backend team was waiting on the NLP team's output format.
- **Action:** I set up a joint working session, got both teams to agree one format and recorded it in the tracker.
- **Result:** The blocker cleared in two days.

3. UAT feedback required rework

- **Situation:** The first UAT cycle produced about 35 feedback items, mostly answers with weak or missing references.
- **Action:** I logged and prioritised every item with the NLP and QA teams and coordinated fixes before the second cycle.
- **Result:** All items were closed before release.

6. Results

The platform went live in production on the planned date.

Measure	Result
Delivery	Production release on the planned date
Sprints	8 two-week sprints
Requirements documented	2 BRDs, 3 FRDs, about 60 requirements
UAT	2 cycles, about 35 feedback items, all closed before release
Answer speed	Under 5 seconds per question

Measure	Result
Search effort for users	About 60% less time finding answers (estimate from client feedback)

The experience strengthened my ability to connect business requirements with engineering execution, maintain visibility across multiple teams and support structured delivery in an AI product environment.

7. Professional Capabilities Demonstrated

- Business and functional requirements management
- BRD and FRD documentation
- Cross-functional project coordination
- Client and stakeholder communication
- Two-week sprint delivery tracking
- Excel-based progress monitoring
- Dependency and blocker follow-up
- Testing and UAT coordination
- Release and production delivery support
- AI/NLP product understanding

Conclusion

This engagement gave me practical experience working at the intersection of business requirements, technical teams and project delivery. It strengthened my ability to coordinate stakeholders, maintain execution visibility, follow up on dependencies and support delivery through production.

I am looking to build on this experience in a Business Analyst role on AI and data products, where I can own requirements end to end and work closely with engineering teams to deliver measurable outcomes.